

Statistical Analysis Report

Analysis Summary Template

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1 Summary

Provide a brief overview of the analysis objectives, key findings, and main conclusions. This section should be concise and accessible to non-technical stakeholders. Max 1 paragraph, no bullet points

2 Introduction

This analysis is to demonstrate the use of Artificial Intelligence (AI) Large Language Models (LLMs) to analyse data. The goal here is to stress test LLMs on a simple analysis task that involves a linear mixed effects model with a binary dependent variable and a blocking factor. The report will be in the form of a statistical analysis report.

3 Data

The data are from a subset of the Chz2006 dataset in the R package {agricolae} (Mendiburu and Yaseen 2020). There are 3 factors of interest (crop, amendment, and block) and a binary dependent variable (wilt_percent). There is also a plant individual factor which we will not consider here. There are $n = 1920$ observations in the data. The data experimentally measures the incidence and performance of healthy tubers and rotten potato field infested with naturally *Ralstonia solanacearum* Race 3/Bv 2A, after application of inorganic amendments and a rotation crop in Carhuaz Peru, 2006.

```
'data.frame': 1920 obs. of 6 variables:
 $ amendment : chr "0C" "0C" "0C" "0C" ...
 $ crop       : chr "Pea" "Pea" "Pea" "Pea" ...
 $ block      : Factor w/ 3 levels "1","2","3": 1 1 1 1 1 1 1 1 1 1 ...
 $ plant      : int 1 2 3 4 5 6 7 8 9 10 ...
 $ wilt_percent: int 0 0 50 0 0 0 0 0 0 0 ...
 $ wilt_bin   : num 0 0 1 0 0 0 0 0 0 0 ...
```

Table 1: Dataset sample

amendment	crop	block	plant	wilt_percent	wilt_bin
0C	Pea	1	1	0	0
0C	Pea	1	2	0	0
0C	Pea	1	3	50	1
0C	Pea	1	4	0	0
0C	Pea	1	5	0	0
0C	Pea	1	6	0	0

4 Statistical Analysis

The statistical analysis consisted of a linear mixed effects model with a binary dependent variable and a blocking factor treating block as a random effect. The model was fitted using the lme4 package in R.

5 Results

The results of the analysis are presented in table 1 with the predictor names and the corresponding p-values, accompanied by a graph of the relationship between each predictor and the response variable.

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: wilt_bin ~ crop + amendment + (1 | block)
Data: wilt
```

AIC	BIC	logLik	-2*log(L)	df.resid
1146.5	1191.0	-565.3	1130.5	1912

Scaled residuals:

Min	1Q	Median	3Q	Max
-0.9244	-0.3003	-0.2264	-0.0968	7.8011

Random effects:

Groups	Name	Variance	Std.Dev.
block	(Intercept)	0.003477	0.05897

Number of obs: 1920, groups: block, 3

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.0396	0.1803	-5.766	8.11e-09 ***
cropCorn	0.2867	0.2291	1.252	0.211
cropFallow	0.8484	0.2170	3.909	9.28e-05 ***
cropPea	0.1879	0.2320	0.810	0.418
amendment3C	-2.1997	0.2254	-9.761	< 2e-16 ***
amendment3C1Z	-1.7068	0.1917	-8.905	< 2e-16 ***
amendment6C	-3.8991	0.4608	-8.462	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

6 References

List any references or citations used in the analysis.

7 Appendix

Session Information

```
sessionInfo()
```

R version 4.5.1 (2025-06-13 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26100)

Matrix products: default
LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United Kingdom.utf8
[2] LC_CTYPE=English_United Kingdom.utf8
[3] LC_MONETARY=English_United Kingdom.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United Kingdom.utf8

time zone: Europe/London
tzcode source: internal

attached base packages:
[1] stats graphics grDevices utils datasets methods base

other attached packages:
[1] sjPlot_2.9.0 lmerTest_3.1-3 lme4_1.1-37 Matrix_1.7-3
[5] knitr_1.50 lubridate_1.9.4 forcats_1.0.1 stringr_1.5.1
[9] dplyr_1.1.4 purrr_1.0.4 readr_2.1.5 tidyr_1.3.1
[13] tibble_3.3.0 ggplot2_3.5.2 tidyverse_2.0.0

loaded via a namespace (and not attached):
[1] generics_0.1.4 stringi_1.8.7 lattice_0.22-7
[4] hms_1.1.3 digest_0.6.37 magrittr_2.0.3
[7] evaluate_1.0.4 grid_4.5.1 timechange_0.3.0
[10] RColorBrewer_1.1-3 fastmap_1.2.0 jsonlite_2.0.0
[13] scales_1.4.0 numDeriv_2016.8-1.1 reformulas_0.4.1
[16] Rdpack_2.6.4 cli_3.6.5 rlang_1.1.6
[19] rbibutils_2.3 splines_4.5.1 withr_3.0.2
[22] yaml_2.3.10 tools_4.5.1 tzdb_0.5.0
[25] nloptr_2.2.1 minqa_1.2.8 boot_1.3-31
[28] vctr_0.6.5 R6_2.6.1 lifecycle_1.0.4
[31] MASS_7.3-65 pkgconfig_2.0.3 pillar_1.10.2
[34] gtable_0.3.6 glue_1.8.0 Rcpp_1.0.14
[37] xfun_0.52 tidysselect_1.2.1 farver_2.1.2
[40] htmltools_0.5.8.1 nlme_3.1-168 rmarkdown_2.29
[43] compiler_4.5.1